- *Sources*—Part of the infrastructure and application layer, such as a microservice, a device, a database, a message queue, or the operating system. They typically must be instrumented to emit signals. We will discuss sources in chapter 3.
- *Agents*—Responsible for signal collection, processing, and routing. Chapter 4 is dedicated to agents and their usage.
- *Destinations*—Where you consume signals, for different reasons and use cases. These include visualizations (e.g., dashboards), alerting, long-term storage (for regulatory purposes), and analytics (finding new usages for an app). We will dive deep into backend and frontend destinations in chapters 5 and 6, respectively.
- *Telemetry*—The process of collecting signals from sources, routing or preprocessing via agents, and ingestion to destinations.

Figure 1.1 provides you with a visual depiction of observability. The motivation is to gather signals from a system represented by a collection of sources via agents to destinations for consumption by either a human or an app, with the goal of understanding and influencing the system.

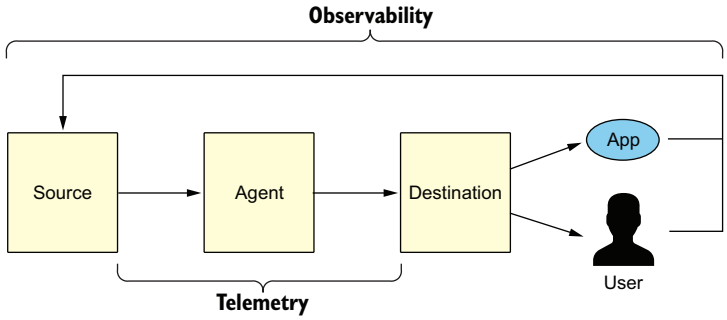


Figure 1.1 Observability overview

Observability represents, in essence, a feedback loop. A human user might, for example, restart a service based on the gathered information. In the case of an app, this could be a cluster autoscaler that adds worker nodes based on the system utilization measured.

The most important aspect of observability is to provide actionable insights. Simply displaying an error message in a log line or having a dashboard with fancy graphics is not sufficient.

> **DEFINITION** *Observability* is the capability to continuously generate and discover actionable insights based on signals from the system under observation, with the goal of influencing the system.

The field of observability is growing and covering more and more domains, including developer observability (which we will cover in chapter 9) and data observability.